JANMITRA -- PHASE 4 (DAYS 31-40) WORK REPORT

Production Push Architecture, Hardened SW, Viva Guide, Report & Capstone Sign-off

Nagpur Municipal Corporation (NMC) Civic Grievance System | Team ResolveIT (SIH25031)

Candidate Information

Name: Mohd Faizan Rayeen (B.Tech Final Year)
Roll No: BE-CSE-2026-042 | Dept of CSE & IT
Role: Citizen Portal Lead & Push Notifications Lead
Sprint Phase: Phase 4 (Days 31 to 40 Final Capstone)

Project & Institution Details

Institution: Priyadarshini Bhagwati College of Engg, Nagpur
Affiliation: Rashtrasant Tukadoji Maharaj Nagpur University
Municipal Partner: Nagpur Municipal Corporation (NMC)
Team ResolveIT: Faizan Rayeen, Anas Shaikh, Anurag Deshmukh, Om Gadekar

Phase 4 Executive Overview & Scope

During Phase 4 (Days 31 to 40), Mohd Faizan Rayeen led the hardening and production deployment of JANMITRA's notification pipeline, testing infrastructure, academic documentation, and project handover package. Key achievements include:

1. Verified Web Push on Production HTTPS with backend token synchronization via PATCH /api/v1/auth/fcm-token/.
2. Hardened Service Worker Caching: Pre-cached static logo/badge PNG assets, added IndexedDB offline sync, and zero-reload hot swapping.
3. Published Complete Testing & Debugging Guide with step-by-step test cases and browser permission reset protocols.
4. Built send_test_notification CLI Command for diagnostic push validation with safe simulation fallback mode.
5. Created Production Testing Checklist validating health endpoints, 25 master demo accounts, and 8-stage complaint lifecycle.
6. Finalized Notification UX in src/firebase.js with gentle floating padlock banner, styled toast with 'View' CTA, and audio chimes.
7. Authored Live Viva Demonstration Guide with rehearsed dual-screen presentation scripts and offline test harness.
8. Wrote Academic Project Report Chapter 1 (Introduction) conforming to IEEE and university capstone standards.
9. Authored 3-Minute Demo Video Walkthrough Script featuring 5 visual callouts and strict real-image sourcing rules.
10. Compiled Final Executive Summary, verified all 11 PDFs in JANMITRA_Final_v3, and created official College Sign-off Document.

Day 1 to Day 40 Full-System Health & Audit Status

Subsystem	Target Files / Endpoints	Verification Standard & Result	Status
Backend Services	manage.py check, models, migrations	0 check errors; 100% migrations applied	**[OK] 100% Pass**
Database Seeder	seed_demo_accounts.py	25 accounts, 12 categories, 21 complaints in DB	**[OK] Seeded**
E2E Journey Test	run_e2e_journey_test.py	5-stage lifecycle (Submitted -> Closed) passed	**[OK] 100% Pass**
Cross-Role Notifs	verify_notification_flow.py	5 stages verified across Citizen, Officer & Worker	**[OK] 100% Pass**
Notification Audit	audit_notifications.py	Background push, toast with View, badge decrement	**[OK] Audited**
Frontend PWA	npm run build, dist/ assets	Compiled in 7.55s; 0 TypeScript/JSX errors	**[OK] Built Clean**
Service Worker	public/firebase-messaging-sw.js	Static cache, IndexedDB sync, zero-reload updates	**[OK] Hardened**
11 Master PDFs	Downloads/JANMITRA_Final_v3	All 11 volumes verified intact and archived	**[OK] Complete**

Days 31 & 32: Production Push Verification & Hardened SW Caching

Live FCM push validation on HTTPS and hardened Service Worker offline architecture

Day 31

Verify FCM Push on Production HTTPS (PRODUCTION_NOTIFICATION_AUDIT.md)

* Problem Solved: Needed live verification of Web Push notifications over secure HTTPS hosting and backend token persistence.
* Production HTTPS Deployment: Configured frontend on Render/Vercel with valid SSL/TLS certificates.
* Token Synchronization: Verified that browser FCM device token automatically registers to Django backend via PATCH /api/v1/auth/fcm-token/.
* Push Delivery Verification: Dispatched test notifications and confirmed native OS desktop banner arrives within 3 seconds.
* Deep-Link Navigation: Verified clicking notification focuses open browser window and navigates directly to target complaint URL.
* Comprehensive Audit Documentation: Documented all 4 test stages, token storage schemas, and production checklist.
* Output File: Created PRODUCTION_NOTIFICATION_AUDIT.md.

Day 32

Harden Service Worker Caching (public/firebase-messaging-sw.js)

* Problem Solved: Standard service workers fail when offline, fail to cache branding icons, and get stuck in waiting state.
* 3 Core Hardening Enhancements:
  1. Static Asset Pre-Caching: Cached /icons/janmitra-logo-192.png and /icons/badge-72.png in 'janmitra-static-v3.2.0-day32'. Added cache-first fetch handler with network fallback to ensure branding icons always display even offline.
  2. Offline Background Sync Fallback: Implemented IndexedDB queue ('janmitra_offline_sync_db' / 'pending_notifications'). When device is offline during push receipt, payload buffers to IndexedDB. Upon reconnect, 'sync' event replays pending alerts.
  3. Zero-Reload Hot Swapping: Added self.skipWaiting() on install and self.clients.claim() on activate to eliminate page reloads.
* Physical Assets Generated: Created 192x192 RGBA logo and 72x72 high-contrast notification badge in public/icons/ and dist/icons/.
* Output File: Updated public/firebase-messaging-sw.js (351 lines, production hardened).

Zero-Reload Service Worker & Offline Sync Architecture

By integrating self.skipWaiting() in the install handler and self.clients.claim() in the activate handler, new Service Worker revisions take control of existing client tabs immediately without interrupting the user. Concurrently, IndexedDB queueing guarantees that offline push notifications are never lost, replaying smoothly upon network re-establishment.

Days 33 & 34: Notification Testing Guide & send_test_notification CLI

Comprehensive debugging guide and automated diagnostic dispatch command

Day 33 — Notification System Tests & Debugging Guide (notification_test_guide.md)

* Problem Solved: Needed a structured, repeatable QA test guide and troubleshooting runbook for the entire notification pipeline.
* 4 Step-by-Step Notification Test Scenarios Documented:
  1. Test 1 (Submitted): Citizen files issue -> Navy toast banner appears with 'View' button + bell badge increments by +1.
  2. Test 2 (Validated): Officer approves complaint -> Citizen receives validation status toast and amber status badge.
  3. Test 3 (Assigned): Officer allocates worker -> Field worker receives audible alert and direct task card link.
  4. Test 4 (Background): All tabs closed -> W3C Service Worker displays OS native banner with logo, badge, and deep-link click.
* Common FCM Issues & Detailed Fixes: VAPID key missing/invalid, Service Worker root scope registration, browser notification permission reset guide (Chrome, Edge, Firefox, Brave, Mobile), and token sync troubleshooting (PATCH /api/v1/auth/fcm-token/).
* Output File: Created notification_test_guide.md (422 lines, synchronized in backend & frontend).

Day 34 — Build send_test_notification Command (send_test_notification.py)

* Problem Solved: Needed a standalone CLI management command to test FCM push delivery without manual database edits.
* 4 Key Engineering Requirements Implemented:
  1. CLI Parameters: Accepts required --user-id=<UUID/ID>, optional --title, --body, --complaint-id, and --dry-run.
  2. User & Token Validation: Queries User by PK/username/email; inspects user.fcm_token and persistent store; provides diagnostic advice.
  3. Firebase Admin SDK Test: Tests SDK initialization; if credentials unavailable, seamlessly activates Safe Simulation Mode (0 crash).
  4. Dispatch & In-App Storage: Formats FCM push payload with click action (/citizen/complaint/:id), resolves Complaint FK, and creates an unread Notification database record to trigger the web NotificationBell.
* Verification: Tested with registered tokens, missing tokens (safe fallback), dry-run mode, and invalid user error handling.
* Output File: Created notifications/management/commands/send_test_notification.py.

Automated CLI Diagnostic Capabilities

The send_test_notification command serves as a critical DevOps diagnostic tool. Operators can verify token health, test payload formats, inspect database foreign-key linkage, and simulate push delivery for QA evaluations even when external network access is restricted.

Days 35 & 36: Production Testing Checklist & Notification UX Finalization

Pre-deployment verification checklist and enhanced client-side notification aesthetics

Day 35

Production Testing Checklist (PRODUCTION_TEST_CHECKLIST.md)

* Problem Solved: Needed a formal pre-flight QA checklist before municipal deployment covering all system components.
* 4 Major Checklists Included:
  1. Health Check & Database: API status (GET /api/v1/health/), Django checks, migrations, and database record audit (25 accounts, 12 categories, 21 complaints).
  2. Multi-Role Auth Checklist: Validated login, redirection, JWT storage, portal UI, and route guards for Citizen, Worker, Officer, and Super Admin.
  3. Complaint Lifecycle: 8-stage verification tracking complaints from Submitted to Closed with citizen feedback rating.
  4. Complete Demo Accounts Reference Table: Full catalog of all 25 accounts (Super Admin 9900000000, 4 Officers 9900000001-4, 16 Field Workers 9900001101-1404, 4 Citizens 9900002001-4) with designations and credentials.
* Output File: Created PRODUCTION_TEST_CHECKLIST.md.

Day 36

Finalise Notification UX (src/firebase.js)

* Problem Solved: Needed elegant UI components for handling denied browser permissions and interactive foreground alerts.
* 3 Core UX Enhancements Implemented:
  1. Gentle Blocked Permission Banner: When Notification.permission === 'denied', renders an amber floating card explaining how to click the browser padlock icon ([Auth]) to set notifications to 'Allow', with dismissal caching and 'Check Again' button.
  2. Styled Foreground Toast: Navy gradient card with NMC Alert badge, prominent Title, Body, Complaint ID tag, and interactive 'View ->' CTA.
  3. Progress Bar & Sensory Feedback: 6-second countdown bar with hover-pause, Web Audio API two-tone chime, and haptic vibration.
* Active/Inactive Receipt Test Harness: Built window.janmitraFCM in DevTools for QA simulation of foreground and background alerts.
* Output File: Updated src/firebase.js; verified production build cleanly via npm run build.

Citizen-First Notification Aesthetics & Permission Handling

Rather than failing silently when browser permissions are blocked, the gentle padlock prompt educates citizens on restoring alerts without intrusive popups. Combined with animated foreground toasts and Web Audio synthesis, the citizen experience feels polished, modern, and trustworthy.

Days 37 & 38: Live Viva Demo Guide & Report Chapter 1 (Introduction)

Dual-screen live viva script and university project report introductory chapter

Day 37 **Push Notification Live Demo Guide (NOTIFICATION_DEMO_GUIDE.md)**

* Problem Solved: Needed a synchronized dual-screen live demonstration procedure and offline contingency plan for project viva.
* Rehearsed Dual-Screen Viva Procedure:
  - Screen A (Phone): Citizen submits civic issue with photo and GPS at WHC Road, Dharampeth.
  - Screen B (Laptop): PWD Officer workstation receives real-time validation toast banner and chime.
  - Officer assigns Ward 2 worker Nitin Bawankar -> Phone sounds chime and displays 'New Task Assigned' alert.
  - Background OS Test: Tab closed -> OS desktop banner arrives with deep-link click handling.
* Offline Simulation Backups: Documented 3 zero-risk fallbacks (Safe Simulation Mode, CLI dispatcher, and DevTools console trigger).
* Examiner Q&A Guide: Authored model answers for background push, offline sync, duplicate suppression, and blocked permissions.
* Output File: Created NOTIFICATION_DEMO_GUIDE.md.

Day 38 **Report Chapter 1: Introduction (REPORT_CHAPTER_1_INTRODUCTION.md)**

* Problem Solved: Needed the formal introductory academic chapter for the final engineering project report.
* 5 Core Sections Authored to Academic / IEEE Standards:
  1. ULB Background: 74th Constitutional Amendment Act (1992), urban decentralization, and Nagpur Municipal Corporation context.
  2. Problem Statement: Absence of automated tracking, black-box delays, manual triage, proof-of-work deficits, and ghost closures.
  3. Project Objectives: Instant reporting, AI validation, dynamic SLA-driven routing, proof-of-work verification, and analytics.
  4. Scope & Limitations: 4 Departments (PWD, SWM, PHE, Electrical), 12 categories, 4 Wards (W1 Laxmi Nagar, W2 Dharampeth, W3 Hanuman Nagar, W4 Mangalwari), 4 RBAC roles, and environmental lighting constraints.
  5. Organization of Report: Roadmap outlining Chapters 1 through 6.
* Output File: Created REPORT_CHAPTER_1_INTRODUCTION.md (22,192 bytes, academic quality).

Academic Rigor & Examination Readiness

Chapter 1 bridges technical software engineering with municipal policy frameworks. By formulating clear problem statements rooted in Nagpur's urban context, the project provides examiners with an academically sound, research-grounded justification for JANMITRA's digital grievance architecture.

Days 39 & 40: 3-Minute Video Script & Final Handover Package

Video walkthrough script, 11 PDF verification, and institutional sign-off certificate

Day 39

Demo Video Walkthrough Script (DEMO_VIDEO_SCRIPT.md)

* Problem Solved: Needed a high-impact, scene-by-scene script for a 3-minute video presentation with strict image authenticity.
* 6-Scene Structured Video Script (0:00 to 3:00):
  - Scene 1 (0:00-0:25): Introduction & Nagpur civic problem context.
  - Scene 2 (0:25-0:55): Citizen mobile submission & [CALLOUT 1: 100m DUPLICATE FLAG].
  - Scene 3 (0:55-1:25): AI computer vision triage & [CALLOUT 2: AI FAKE CHECK].
  - Scene 4 (1:25-1:55): Department Officer dashboard & [CALLOUT 3: PUSH NOTIFICATION TOAST].
  - Scene 5 (1:55-2:30): Field worker on-site execution & [CALLOUT 4: PROOF OF WORK].
  - Scene 6 (2:30-3:00): [CALLOUT 5: 6-STEP STEPPER], citizen 5-star rating, and NMC impact.
* Strict Image Authenticity Policy: Mandated real photographs of Nagpur infrastructure (WHC Road, Great Nag Road); strictly prohibited AI-generated images because Anurag's computer vision model evaluates authentic sensor captures.
* Output File: Created DEMO_VIDEO_SCRIPT.md.

Day 40

Final Project Handover Package & Sign-Off

* Problem Solved: Needed final capstone handover compilation, documentation archive verification, and college sign-off.
* 3 Core Deliverables Completed:
  1. Final Executive Summary (FINAL_EXECUTIVE_SUMMARY.md): Comprehensive document covering all 40 days of engineering.
  2. 11 Documentation Volumes Verified: Confirmed all 11 PDFs present in JANMITRA_Final_v3 (01_PRD, 02_SRS, 03_Architecture, 04_UIUX, 05_Development, 06_Team, 07_Dept_Dashboard, 08_Worker_Management, 09_TechStack, 10_Project_Breakdown, 11_Prompts).
  3. Project Completion Sign-Off Certificate (PROJECT_COMPLETION_SIGNOFF.md): Formal academic completion certificate for Priyadarshini Bhagwati College of Engineering (PBCOE), Nagpur and RTMNU with signature blocks for Guide, HOD, and Examiner.
* Team Continuation Note: Faizan's modules (Citizen Portal & Push Notifications) are 100% complete; seamlessly interfacing with Anas Shaikh (Backend Lead) and Anurag Deshmukh (AI/ML Lead) as they finalize their remaining components.
* Output Files: Created FINAL_EXECUTIVE_SUMMARY.md and PROJECT_COMPLETION_SIGNOFF.md.

Capstone Milestone: 100% 40-Day Trajectory Completed

With Day 40 completed, the JANMITRA platform stands fully verified, documented, and packaged. From initial product requirements to production Service Worker caching and live viva demonstration scripts, the system represents an exemplary final-year engineering achievement for Priyadarshini Bhagwati College of Engineering and the Nagpur Municipal Corporation.

Master Documentation Archive & College Sign-Off Summary

Verification matrix of all 11 compiled volumes in JANMITRA_Final_v3 directory

Vol	Document Name	Size	Focus Area & Scope	Status
01	01_PRD_JANMITRA_v3.pdf	14.8 KB	Product Requirements, User Personas, NMC Vision	[v] Verified
02	02_SRS_JANMITRA_v3.pdf	16.8 KB	IEEE Functional & Non-Functional Requirements	[v] Verified
03	03_Architecture_JANMITRA_v3.pdf	9.6 KB	3-Tier System Architecture & Data Flow Models	[v] Verified
04	04_UIUX_JANMITRA_v3.pdf	11.2 KB	Design System, Glassmorphism, Color Palette	[v] Verified
05	05_Development_JANMITRA_v3.pdf	19.9 KB	Dev Setup, Docker Configs, REST Endpoints	[v] Verified
06	06_Team_JANMITRA_v3.pdf	9.4 KB	Role Matrix, Team Execution Plan (ResolveIT)	[v] Verified
07	07_Dept_Dashboard_JANMITRA_v3.pdf	12.1 KB	Departmental Triage (PWD, SWM, PHE, ELE)	[v] Verified
08	08_Worker_Management_JANMITRA_v3.pdf	11.4 KB	Field Worker Workflows & Proof of Work	[v] Verified
09	09_TechStack_Consistency_JANMITRA_v3.pdf	15.6 KB	Dependencies, Security & Runtime Standards	[v] Verified
10	10_Project_Breakdown_JANMITRA.pdf	15.3 KB	Work Breakdown Structure (40-Day Sprint)	[v] Verified
11	11_Day_by_Day_Prompts_JANMITRA.pdf	171.0 KB	Complete 40-Day Prompt Engineering Trajectory	[v] Verified

PRIYADARSHINI BHAGWATI COLLEGE OF ENGINEERING, NAGPUR -- PROJECT SIGN-OFF

Department of Computer Science & Engineering | RTM Nagpur University

This is to certify that candidate Mohd Faizan Rayeen (Roll No: BE-CSE-2026-042) has successfully completed all assigned engineering milestones for the major capstone project JANMITRA across Days 1 through 40. The software systems, push notification architectures, and technical documentation have been audited and found to satisfy all requirements for B.Tech degree evaluation.

Mohd Faizan Rayeen
Lead Presenter

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